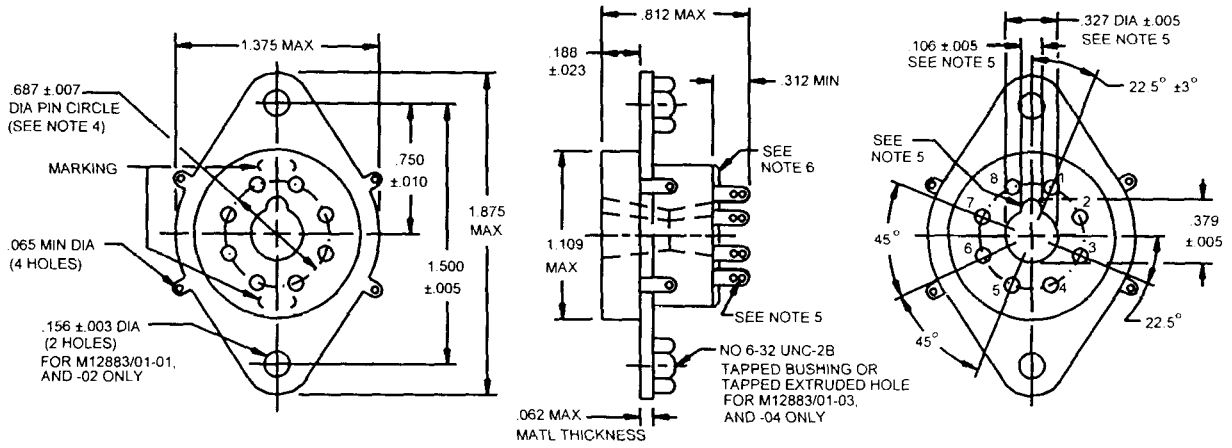
INCH-POUND
MIL-PRF-12883/1E
28 June 1996
SUPERSEDING
MIL-S-12883/1D
11 January 1972

PERFORMANCE SPECIFICATION SHEET

SOCKETS AND ACCESSORIES FOR PLUG-IN ELECTRONIC COMPONENTS (ELECTRON TUBE, BOTTOM MOUNTING, SADDLE TYPE 8 CONTACT, RADIAL)

This specification is approved for use by all Departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-12883.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.062	1.57	.312	7.92	.812	20.62
.005	0.13	.065	1.65	.327	8.31	1.109	28.17
.007	0.18	.106	2.69	.379	9.63	1.375	34.93
.010	0.25	.156	3.96	.687	17.45	1.500	38.10
.023	0.58	.188	4.78	.750	19.05	1.875	47.63

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.016 inch (0.41 mm) and ±1/2° on angles.
4. Eight contact cavities equally spaced, and each cavity located within 1/2° of true position, shall be established along the pin circle.
5. Each contact tab shall have either 2 wire holes of .065 inch (1.65 mm) minimum diameter, or 1 hole of .125 inch (2.39 mm) minimum length and .065 inch (1.65 mm) minimum width. The hole, or holes, shall lie on the longitudinal centerline of the contact tab within plus or minus .008 inch (0.20 mm).
6. Barriers and their shapes are optional. When present, barriers shall have a maximum height of .094 inch (2.39 mm).
7. Keyway may be square or rounded. For rounded keyways, radius shall be .053 inch (1.35 mm), and blending shall be allowed at the junction of the keyway and the center hole. Center-pin hole shall be relieved to allow the locating pin freedom to rock when a tube is withdrawn. Method of meeting this requirement is optional.
8. Material thickness of mounting flange shall be measured in an area where burring or dishing of the mounting hole is not present.

FIGURE 1. Socket configuration.

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REQUIREMENTS:

Dimensions and configurations: See figure 1, and table I.

Reference material: (see 3.1.1 of MIL-PRF-12883).

Insulating material: Ceramic, SDG-F, GDI-30F, or MIL-M-14 type MFE, see table I.

Contact cavities: One shape only; D-shaped or circular, at option of the manufacturer.

Electrical

Insulation resistance: 1,000 megohms minimum. Test pin diameter: .093 ±.001 inch (2.36 ±.03 mm).

Dielectric withstanding voltage:

Sea level: Test voltage: 3,500 volts root mean square (rms). Test pin diameter: .093 ±.001 inch (2.36 ±.03 mm).

High altitude: Test voltage: 1,000 volts rms. Test pin diameter: .093 ±.001 inch (2.36 ±.03 mm).

Contact resistance:

Average for all contacts: 0.015 ohm maximum.

Individual contacts: 0.03 ohm maximum.

Continuity test circuit: The header shall be the one used for electron tube 6AC7WA.

Mechanical

Insertion and withdrawal force:

Initial insertion force: 25 pounds (111.21 newton) maximum.

Initial withdrawal force: 8 pounds (35.59 newton) minimum, 20 pounds (88.96 newton) maximum.

Torque: M12883/01-03 and -04 sockets shall be subjected to a torque of 12 pounds-inches (53.38 newton).

Vibration: The test gage shall be in accordance with MIL-PRF-12883, appendix, figure 3.

Durability: After the durability test, the withdrawal force shall be 5 pounds (22.24 newton) minimum.

Static load: 80 pounds (355.86 newton).

The part shall be designed to operate at the following conditions:

Environmental

Operating temperature: See table I.

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TABLE I. Dash number and characteristics.

Dash number	Operating temperature °C	Insulating material	Mounting style	Old type designator
-01	200	Ceramic	Clearance hole	TS101C01
-02	100	SDG-F, GDI-30, or MIL-M-14 (type MFE)	Clearance hole	TS101P01
-03	200	Ceramic	Threaded bushing	TS101C02
-04	100	SDG-F, GDI-30, or MIL-M-14 (type MFE)	Threaded bushing	TS101P02

Ratings (absolute maximum):

Voltage:
Sea level: 1,250 volts rms.
50,000 feet (15240 meters): 850 volts rms.

Current: 5 amperes.

Operating temperature: See table I.

Test gage details: See table II and MIL-PRF-12883 appendix.

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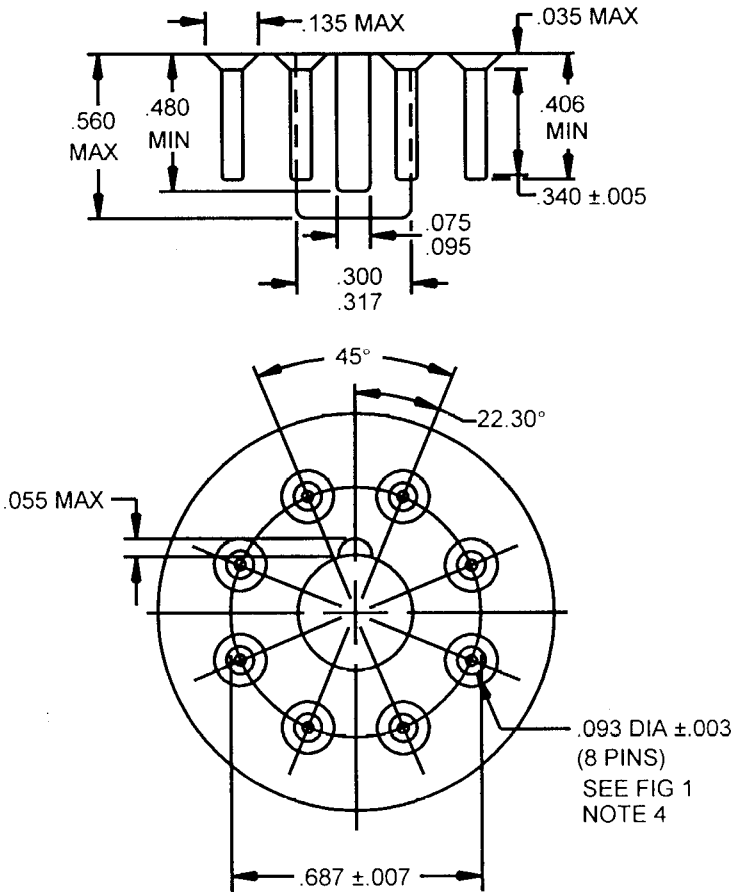
TABLE II. Test gage details. 1/ 2/

Inspection	A Pin length (mm)	B Pin dia (mm)	M Test-end dia (mm)	C Pin-circle dia (mm)	N Probe-end dia (mm)	D Gage dia max (mm)	No. of pins	Total weight of gage ounces (grams) (±5%)
Insertion and withdrawal force------	.437 ±.005 (11.10 ±0.13)	.0905 ±.0001 (2.999 ±0.003)	---	.6870 (17.450) basic	---	1-7/64 (28.18)	8	---
Contact resistance-	.437 ±.005 (11.10 ±0.13)	.0905 ±.0001 (2.999 ±0.003)	---	.6870 (17.450) basic	---	1-7/64 (28.18)	8	---
Contact retention--	.437 ±.005 (11.10 ±0.13)	---	.0905 ±.0001 (2.999 ±0.003)	---	.0955 ±.0001 (2.426 ±0.003)	---	---	2 (56.7)
Durability---	.437 ±.005 (11.10 ±0.13)	.0955 ±.0001 (2.426 ±0.003)	---	.6870 (17.450) basic	---	1-7/64 (28.18)	8	---

1/ Dimensions are in inches.
2/ Metric equivalents are given for general information only.

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Mating-base dimensions: Sockets shall accommodate plug-in components having mating-base dimensions as shown on figure 2.



Inch	mm
.003	0.08
.005	0.13
.007	0.18
.035	0.89
.055	1.40
.075	1.91
.093	2.36
.095	2.41
.135	3.43
.300	7.62
.317	7.66
.340	8.64
.406	10.31
.480	12.19
.560	14.22
.687	17.45

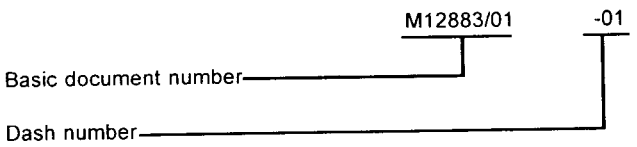
NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Mating-base dimensions.

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Part or identifying number (PIN): The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example



NOTE:

This specification is subject to NATO international standardization agreement NEPR 67.

The Government PIN, specified in table III, supersedes the following commercial part numbers.

TABLE III. Supersession and cross reference data.

Active Government PIN	Superseded part number					
	CAGE 72825	CAGE 32657	CAGE 49956	CAGE 08385	CAGE 56232	CAGE 06481
M12883/01-01 M12883/01-02 M12883/01-03 M12883/01-04	TBD 9751-77 TBD TBD	R883B15 2 NA NA NA	282-1043P2 NA NA NA	305001572 NA NA NA	752854-002 NA NA NA	900999-6662 NA NA NA

TABLE III. Supersession and cross reference data - Continued.

Active Government PIN	Superseded part number		
	CAGE 02734	CAGE 49671	CAGE H0203
M12883/01-01 M12883/01-02 M12883/01-03 M12883/01-04	99393-2 NA NA NA	99393-2 NA NA NA	35225002816 6 NA NA NA

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CONCLUDING MATERIAL

Custodian:
Navy - EC
ARMY - CR
Air Force - 85

Review activity:
Navy - AS, CG, MC, SH
Air Force - 11, 17, 99
Army - AT, AV, EA, MI, ME

Preparing activity:
DLA - ES

(Project 5935-4015-01)

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